Pushkar Dave

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Education

Northwestern University Evanston, IL
Master of Science in Robotics *December 2025*

Visvesvaraya National Institute of Technology Nagpur, INDIA
Bachelor of Technology in Electrical Engineering *May 2024*

Skills

Software: Git, Linux, C++, Python, C, RTOS, Docker, Bash, MATLAB, Protobuf, CMake, Unit Testing
Robotics: ROS2, ROS, SLAM, Gazebo, RViz, OpenCV, MoveIt, Nav2, PX4, Robot Kinematics, Control Systems, Computer Vision
Hardware: Onshape, PCB Design, KiCad, Embedded Systems, I2C, UART, SPI, RS485, Oscilloscopes, Logic Analyzer
Machine Learning: PyTorch, Reinforcement Learning, Deep Learning, Transformers, Autoencoders, CNNs, MuJoCo

Work Experience

Robotics Engineer March 2026 – Present
PSYONIC Inc. *San Diego, CA*
- Engineering a bimanual teleoperation system integrating dexterous hands, motion-capture gloves, and a multi-DOF haptic device for real-time manipulation of shipping locks
- Designed a KD-tree-based retargeting pipeline mapping mocap glove data to finger commands under kinematic uncertainty
- Built multithreaded message-routing executables to parse and direct sensor data to downstream devices with minimal latency
- Implemented low-latency UDP communication between ABB robots and the teleoperation system using Protobuf messaging
- Contributed to a firmware control wrapper exposing validated control modes with checksum validation, reply parsing, and timeout handling

Robotics Integration Intern June 2025 - August 2025
Caterpillar Inc. *Decatur, IL*
- Automated visual weld inspection by integrating a laser scanner with classical CV-based feature detection
- Collected and curated a 30+ sample weld dataset via cobot, identifying failure modes and edge cases
- Improved scan quality by tuning scan rate and resolution to match cobot velocity, resolving sampling inconsistencies

Robotics Research Intern May 2023 - September 2023
Multi-robot Systems Group, Czech Technical Univeristy *Prague, CZECHIA*
- Developed a triangulation algorithm in C++ to correct vertical drift for the leader UAV in a swarm system
- Performed localization using follower UAVs by fusing UVDAR, IMU, RangeFinder data with a Kalman Filter
- Analyzed and recorded ROS simulation metrics and debugged plots to set up real world experiments

Robotics Intern July 2021 – October 2021
IvLabs, VNIT *Nagpur, INDIA*
- Implemented PD control system and minimum snap trajectory generation for quadrotors in MATLAB
- Designed a state space quadrotor model and solved seventh-order polynomial functions for trajectories
- Experimentally modeled a tethered quadrotor by modeling it as a damped spring-mass system

Featured Projects

Emergent Locomotion in Handed Shearing Auxetic (HSA) Actuated Robot April 2025 – December 2025
- Modeled HSA actuators in MuJoCo using spring–motor systems to simulate torsional forces driving extension & contraction
- Built a Python control interface for the soft robots, enabling stateful rolling and crawling behaviors in simulation
- Integrated the Proximal Policy Optimization RL algorithm, to train locomotion models over varied terrain conditions

Collaborative Mapping using a Quadruped and Quadrotor January 2025 – March 2025
- Created occupancy grids using ORB feature extraction, FLANN feature mapping, loop closure using RTABMap in C++
- Optimized ROS 2 middleware, enabled data throttling and transport relay to achieve lossless camera streaming
- Deployed multi-session mapping to generate an exhaustive and feature-rich point cloud

Low Level Motor Controller using PIC32 February 2025 - March 2025
- Designed and implemented a DC motor control system in C using PIC32 with dual interrupt architecture
- Programmed firmware featuring PWM signal generation, state machine, encoder communication with UART
- Achieved 98% trajectory tracking using PID tuning running on 5KHz current control and 200Hz position control